Cours MALG & MOVEX

MALG

Vérification mécanisée de contrats (III) (The ANSI/ISO C Specification Language (ACSL))

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Plan

1. Contracts
   - Extending C programming
   - language by contracts
   - Playing with variables
   - Ghost Variables
   - Labels
2. Generation of Verification Conditions
   - WP calculus in Frama-c
   - First annotation
   - Second annotation
3. Memory Models in Frama-c
4. Logic Specification
5. Organisation of the verification process
6. Conclusion

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Outline

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Summary on annotations and assertions

- requires
- assigns
- ensures
- decreases
- predicate
- logic
- lemma

Current Summary

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Programming by contract

- The calling function should garantee the required condition or precondition introduced by the clauses requires $P1 \wedge \ldots \wedge Pn$ at the calling point.
- The called function returns results that are ensured by the clause ensures $E1 \wedge \ldots \wedge Em$; ensures clause exporess a relatrionship between the initial values of variables and the final values.
- initial values of a variable v is denoted $\backslash old(v)$
- The variables which are not in the set $L1 \cup \ldots \cup Lp$ are not modified.

Listing 1 – contrat

```
/*@ requires P1;...;requires Pn;
  @ assigns L1;...;assigns Lm;
  @ ensures E1;...;ensures Ep;
  @*/
```

Examples of contract (1)

(Division)

Listing 2 – project-divers/annotation.c

```
/*@ requires x >= 0 && x <= 10;
  @ assigns \nothing;
  @ ensures x % 2 == 0 ==> 2*\result == x;
 @ ensures x % 2 != 0 ==> 2*\result == x-1;
*/
int f(int x)
{
int y;
y = x/2 ;
return(y);
}
```

Examples of contract (1)

(Division)

Listing 3 – project-divers/annotationwp.c

```
/*@ requires 0 <= x  && x <=  10;
  @ assigns \nothing;
  @ ensures  x % 2 == 0 ==> 2*\result == x;
  @ ensures  x % 2 != 0 ==> 2*\result == x-1;
  @*/

int f(int x)
{
/*@ assert x % 2 == 0 ==> 2* (x / 2) == x; */
/*@ assert   x % 2 != 0 ==> 2* (x / 2) == x-1; */
  int y;
/*@ assert x % 2 == 0 ==> 2* (x / 2) == x; */
/*@ assert   x % 2 != 0 ==> 2* (x / 2) == x-1; */
  y  = x / 2;
/*@ assert x % 2 == 0 ==> 2*y == x; */
/*@ assert   x % 2 != 0 ==> 2*y == x-1; */
  return(y);
/*@ assert x % 2 == 0 ==> 2*y == x; */
/*@ assert   x % 2 != 0 ==> 2*y == x-1; */

  }
```

Examples of contract (1)

Property to check
x is the value of the variable x when calling the function $x \geq 0 \wedge x < 10; \Rightarrow \left(\begin{array}{lcl} x \ \% \ 2 & = & 0 \Rightarrow 2 \cdot (x/2) = x \\ x \ \% \ 2 & \neq & 0 \Rightarrow 2 \cdot (x/2) = x - 1 \end{array} \right)$

Proof Obligations from Frama-c

```
  Goal Post-condition (file annotation.c, line 3) in 'annotation':
Let x_1 = x / 2.
Assume {
  Type: is_sint32(x) /\ is_sint32(x_1).
  (* Goal *)
  When: (x % 2) = 0.
  (* Pre-condition *)
  Have: (0 <= x) /\ (x <= 10).
}
Prove: (2 * x_1) = x.
Prover Alt-Ergo 2.6.0 returns Valid (Qed:0.71ms) (13ms) (29)


Goal Post-condition (file annotation.c, line 4) in 'annotation':
Let x_1 = x / 2.
Assume {
  Type: is_sint32(x) /\ is_sint32(x_1).
  (* Goal *)
  When: (x % 2) != 0.
  (* Pre-condition *)
  Have: (0 <= x) /\ (x <= 10).
}
Prove: (1 + (2 * x_1)) = x.
Prover Alt-Ergo 2.6.0 returns Valid (Qed:0.55ms) (14ms) (37)
}
```

Verification of a contract

A programme P *fulfils* a contract (pre,post) :

- P transforms a variable v à partir d'une valeur initiale v_0 and producing a final value v_f : $x_0 \xrightarrow{P} v_f$
- x_0 satisfait pre : pre(v_0) and v_f satisfait post : post(v_0, v_f)
- pre$(v_0) \wedge v_0 \xrightarrow{P} v_f \Rightarrow$ post(v_0, v_f)

requires $pre(in_0)$
ensures $post(in_0, out_f)$
variables in, out
$type1 \quad f(type2\ in_0)$
local variables x
  begin
    $S(f)$
  $return(out)$;
  end

- $v = (in, out, x)$: in (input parameters), out (results), x (local variables)
- pre$(in_0) \wedge (in_0, out_0, v_0) \xrightarrow{P} (in_0, out_f, v_f) \Rightarrow$ post(in_0, out_f)
- pre$(in_0) \wedge in = in_0 \wedge out = out_0 \Rightarrow [S(f)](post(in_0, out))$

Verification Condition for Assignment

Listing 4 – an1.c

```
//@ assert l1: P(x);
  x = e(x);
//@ assert l2: Q(x);
```

Verification Condition for Assignment

Listing 5 – an1.c

```
//@ assert l1: P(x);
  x = e(x);
//@ assert l2: Q(x);
```

- $P(x) \Rightarrow WP(x := e(x))(Q(x))$

Verification Condition for Assignment

Listing 6 – an1.c

```
//@ assert l1: P(x);
  x = e(x);
//@ assert l2: Q(x);
```

- $P(x) \Rightarrow WP(x := e(x))(Q(x))$
- $P(x) \Rightarrow Q[x \mapsto e(x))]$

Verification Condition for Assignment

Listing 7 – an1.c

```
//@ assert l1: P(x);
  x = e(x);
//@ assert l2: Q(x);
```

- $P(x) \Rightarrow WP(x := e(x))(Q(x))$
- $P(x) \Rightarrow Q[x \mapsto e(x))]$
- $P(x1) \Rightarrow Q[x \mapsto e(x1))]$ (renaming of free occurences of x by x1)

Verification Condition for Assignment

Listing 8 – an1.c

```
//@ assert l1: P(x);
  x = e(x);
//@ assert l2: Q(x);
```

- $P(x) \Rightarrow WP(x := e(x))(Q(x))$
- $P(x) \Rightarrow Q[x \mapsto e(x))]$
- $P(x1) \Rightarrow Q[x \mapsto e(x1))]$ (renaming of free occurences of x by x1)
- $P(x1) \wedge x = e(x1) \Rightarrow Q(x)$

Verification Condition for Assignment

Listing 9 – an1.c

```
//@ assert l1: P(x);
  x = e(x);
//@ assert l2: Q(x);
```

- $P(x) \Rightarrow WP(x := e(x))(Q(x))$
- $P(x) \Rightarrow Q[x \mapsto e(x))]$
- $P(x1) \Rightarrow Q[x \mapsto e(x1))]$ (renaming of free occurences of x by x1)
- $P(x1) \wedge x = e(x1) \Rightarrow Q(x)$
- $P(x1) \wedge x = e(x1) \Rightarrow Q(x)$

Verification Condition for Assignment

Listing 10 – an1.c

```
//@ assert l1: P(x);
  x = e(x);
//@ assert l2: Q(x);
```

- $P(x) \Rightarrow WP(x := e(x))(Q(x))$
- $P(x) \Rightarrow Q[x \mapsto e(x))]$
- $P(x1) \Rightarrow Q[x \mapsto e(x1))]$ (renaming of free occurences of x by x1)
- $P(x1) \wedge x = e(x1) \Rightarrow Q(x)$
- $P(x1) \wedge x = e(x1) \Rightarrow Q(x)$
- $\vdash P(x1) \wedge x = e(x1) \Rightarrow Q(x)$

Verification Condition for Assignment

Listing 11 – an1.c

```
//@ assert l1: P(x);
  x = e(x);
//@ assert l2: Q(x);
```

- $P(x) \Rightarrow WP(x := e(x))(Q(x))$
- $P(x) \Rightarrow Q[x \mapsto e(x))]$
- $P(x1) \Rightarrow Q[x \mapsto e(x1))]$ (renaming of free occurences of x by x1)
- $P(x1) \wedge x = e(x1) \Rightarrow Q(x)$
- $P(x1) \wedge x = e(x1) \Rightarrow Q(x)$
- $\vdash P(x1) \wedge x = e(x1) \Rightarrow Q(x)$
- $P(x1) \wedge x = e(x1) \vdash Q(x)$

Verification Condition for Assignment

Listing 12 – an1.c

```
//@ assert l1: P(x);
  x = e(x);
//@ assert l2: Q(x);
```

- $P(x) \Rightarrow WP(x := e(x))(Q(x))$
- $P(x) \Rightarrow Q[x \mapsto e(x))]$
- $P(x1) \Rightarrow Q[x \mapsto e(x1))]$ (renaming of free occurences of x by x1)
- $P(x1) \wedge x = e(x1) \Rightarrow Q(x)$
- $P(x1) \wedge x = e(x1) \Rightarrow Q(x)$
- $\vdash P(x1) \wedge x = e(x1) \Rightarrow Q(x)$
- $P(x1) \wedge x = e(x1) \vdash Q(x)$

  Assume $\{$
  $P(x1)$
- $x = e(x1)$
  $\}$
  Prove:$Q(x)$
-

Verification conditions

Verification conditions

```
  Goal Assertion
  (file annotations14.c, line 5):
Assume {
  Type: is_sint32(x) /\ is_sint32(x_1).
  Have: x_1 = 1.
  Have: (1 + x_1) = x.
}
Prove: x = 2.
Prover Qed returns Valid
```

Verification conditions

```
  Goal Assertion
  (file annotations14.c, line 5):
Assume {
  Type: is_sint32(x) /\ is_sint32(x_1).
  Have: x_1 = 1.
  Have: (1 + x_1) = x.
}
Prove: x = 2.
Prover Qed returns Valid
```

```
void main(void){
  int x;
  x = 1;
  x = x + 1;
  /*@  assert x == 2; */

  }
```

Verification conditions

```
  Goal Assertion
  (file annotations14.c, line 5):
Assume {
  Type: is_sint32(x) /\ is_sint32(x_1).
  Have: x_1 = 1.
  Have: (1 + x_1) = x.
}
Prove: x = 2.
Prover Qed returns Valid
```

```
void main(void){
  int x;
  x = 1;
  x = x + 1;
  /*@  assert x == 2; */

  }
```

```
        Goal Assertion
        (file annotations15.c, line 7):
  Assume {
    Type: is_sint32(x) /\ is_sint32(x_1)
         /\ is_sint32(x_2) /\ is_sint32(y).
    Have: x_2 = 1.
    Have: (1 + x_2) = x_1.
    Have: (2 * x_1) = y.
    Have: (x_1 + y) = x.
  }
  Prove: x = 6.
  Prover Qed returns Valid
```

Verification conditions

```
  Goal Assertion
  (file annotations14.c, line 5):
Assume {
  Type: is_sint32(x) /\ is_sint32(x_1).
  Have: x_1 = 1.
  Have: (1 + x_1) = x.
}
Prove: x = 2.
Prover Qed returns Valid
```

```
void main(void){
  int x;
  x = 1;
  x = x + 1;
  /*@  assert x == 2; */

  }
```

```
        Goal Assertion
        (file annotations15.c, line 7):
  Assume {
    Type: is_sint32(x) /\ is_sint32(x_1)
         /\ is_sint32(x_2) /\ is_sint32(y).
    Have: x_2 = 1.
    Have: (1 + x_2) = x_1.
    Have: (2 * x_1) = y.
    Have: (x_1 + y) = x.
  }
  Prove: x = 6.
  Prover Qed returns Valid
```

```
void main(void){
  int x,y;
  x = 1;
  x = x + 1;
  y = 2*x;
  x = x+y;
  /*@  assert x == 6; */

  }
```

Examples of contract (2)

(Precondition)

Listing 13 – project-divers/annotation0.c

```
/*@ requires x >= 0 && x <  0;
  @ assigns \nothing;
  @ ensures \result == 0;
  @*/
int annotation0(int x)
{
  int y;
  y  =  y / (x-x);
  return(y);
  }
```

Examples of contract (2)

(Precondition)

Listing 14 – project-divers/annotation0wp.c

```
/*@ requires x >= 0 && x <  0;
  @ assigns \nothing;
  @ ensures \result == 0;
  @*/
int annotation(int x)
{
  /*@ assert y / (x-x) == 0; */
  int y;
  /*@ assert y / (x-x) == 0; */
  y  =  y / (x-x);
  /*@ assert y == 0; */
  return(y);
  /*@ assert y == 0; */
  }
```

Examples of contract (2)

Property to check

$0 \leq x \wedge x \leq 10 \Rightarrow y/(x-x) = 0$

Definition of a contract (specification)

- Define the mathematical fucntion to compute (what to compute ?)
- Define an inductive method for computing the mathematical function and using axioms.

(facctorial what)

Listing 15 – project-factorial/factorial.h

```
#ifndef _A_H
#define _A_H
/*@ axiomatic mathfact {
  @ logic integer mathfact(integer n);
  @ axiom mathfact_1: mathfact(1) == 1;
  @ axiom mathfact_rec: \forall integer n; n > 1
  ==> mathfact(n) == n * mathfact(n-1);
  @ } */

/*@ requires n > 0;
  decreases n;
  ensures \result == mathfact(n);
  assigns \nothing;
*/
int codefact(int n);
#endif
```

Definition of a contract (programming)

- Define the program codefact for computing mathfact (How to compute ?)
- Define the algorithm computing the function mathfact

(facctorial how)

Listing 16 – project-factorial/factorial.c

```
#include "factorial.h"

int codefact(int n) {
  int y = 1;
  int x = n;
  /*@ loop invariant x >= 1 && x <= n && mathfact(n) == y * mathfact(x);
    loop assigns x, y;
    loop variant x;
   */
  while (x != 1) {
    y = y * x;
    x = x - 1;
  };
  return y;

}
```

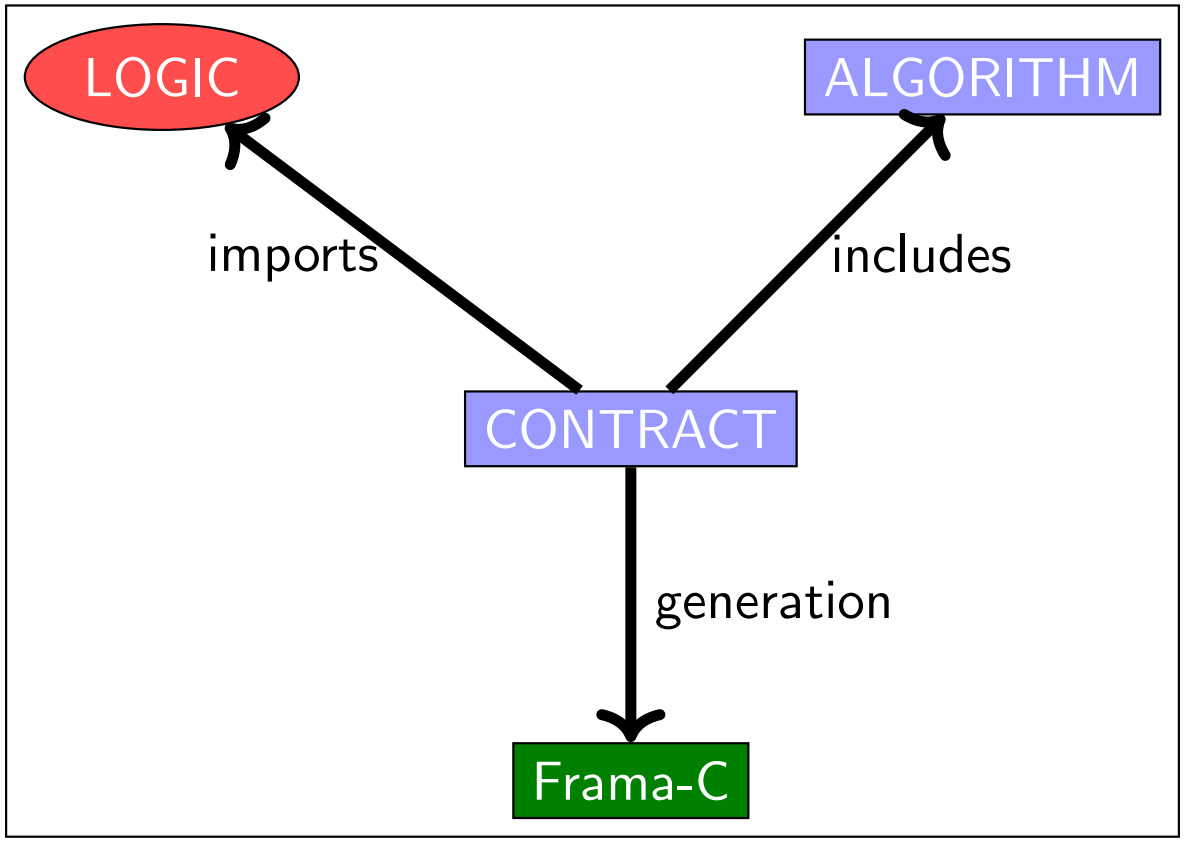
Definition of a contract (approach)

- The specification of a function (mathfact) to compute requires to define it mathematically.
- The definition is stated in an axtiomatic framework and is preferably inductive (mathfact) which is used in assrtions or theorems or lemmas.
- The relationship between the ciomputed value (\result) and the mathematical value (mathfact(n)) is stated in the ensures clause :

$$\backslash\text{result} == \text{mathfact}(n)$$

- The main property to prove is codefact(n)==mathfact(n) : Calling codefact for n returns a value equal to mathfact(n).

Tryptic Pattern



Contracts with named behaviours

Listing 17 – contrat

```
/*@ requires P;
@ behavior b1:
  @ assumes A1;
  @ requires R1 ;
  @ assigns L1;
  @ ensures E1;
@ behavior b2:
  @ assumes A2;
  @ requires R2;
  @ assigns L2;
  @ ensures E2;
@*/
```

Division should not return silly expressions !

(Pairs of integers)

Listing 18 – project-divers/structures.h

```
#ifndef _STRUCTURE_H


struct s {
    int q;
    int r;
};

#endif
```

Division should not return silly expressions !

(Specification)

Listing 19 – project-divers/division.h

```
#ifndef _A_H
#define _A_H
#include "structures.h"
/*@ requires a >= 0 && b >= 0;
 @ requires b != 0;
  @ ensures 0 <= \result.r;
  @ ensures \result.r < b;
  @ ensures  a ==  b * \result.q + \result.r;
*/
struct s  division(int a, int b);
#endif
```

Division should not return silly expressions !

(Algorithm)

Listing 20 – project-divers/division.c

```
#include <stdio.h>
#include <stdlib.h>
#include "division.h"

struct s  division(int a, int b)
{  int rr = a;
   int qq = 0;
   struct s  silly = {-1,-1};
   struct s resu;
   if (b == 0) {
     return silly;
   }
   else
     {
  /*@
    loop invariant
    ( a ==  b*qq + rr) &&
    rr >= 0;
    loop assigns rr,qq;
    loop variant rr;
   */
   while (rr >= b) { rr = rr - b; qq=qq+1;};
   resu.q= qq;
   resu.r = rr;
  return resu;
}
}
```

Iteration Rule for PC

If $\{P \wedge B\}\mathbf{S}\{P\}$, then $\{P\}$**while B do S od**$\{P \wedge \neg B\}$.

- Prove $\{P \wedge B\}\mathbf{S}\{P\}$ or $P \wedge B \Rightarrow \{\mathbf{S}\}(P)$.
- By the iteration rule, we conclude that $\{P\}$**while B do S od**$\{P \wedge \neg B\}$ without using WLP.
- Introduction of LOOP INVARIANTS in the notation.

Listing 21 – loop.c

```
/*@ loop invariant I1;
    loop invariant I2;
    ...
    loop invariant In;
    loop assigns X;
    loop variant E;
  */
```

(Invariant de boucle)

Listing 22 – project-divers/anno6.c

```
/*@ requires a >= 0 && b >= 0;
  ensures 0 <= \result;
  ensures \result < b;
  ensures \exists integer k; a == k * b + \result;
*/
int rem(int a, int b) {
  int r = a;
  /*@
    loop invariant
    (\exists integer i; a == i * b + r) &&
    r >= 0;
    loop assigns r;
   */
  while (r >= b) { r = r - b;};
  return r;
}
```

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Initial value of x $\backslash old(x)$

- $\backslash old(x)$ is the value of the variable when the function is called.
- It can be used in the postcondition of the *ensures* clause.

(Modifying variables while calling)

Listing 23 – project-divers/old1.c

```
/*@ requires \valid(a) && \valid(b);
   @ assigns *a,*b;
    @ ensures    *a == \at(*a,Pre) +2;
    @ ensures  *b == \at(*b,Pre)+\at(*a,Pre)+2;

        @ ensures  \result == 0;
*/
int old(int *a, int *b) {
  int x,y;
  x = *a;
  y = *b;
  x=x+2;
   y = y +x;

  *a = x;
  *b = y;
  return 0 ;
}
```

Expression $\backslash at(e, id)$

- $\backslash at(e, id)$ is the value of e at the control point id.
- id should occur before $\backslash at(e, id)$
- id is one of the possible expressions : Pre, Here, Old, Post, LoopEntry, LoopCurrent, Init
- $\backslash old(e)$ is equivalent to $\backslash at(e, Old)$

Exemple pour $\backslash at(e, id)$

(label Pre)

Listing 24 – project-divers/at1.c

```
/*@
  requires \valid(a) && \valid(b);
  assigns *a,*b;
  ensures  *a == \old(*a)+2;
  ensures  *b == \old(*b)+\old(*a)+2;
*/
int at1(int *a, int *b) {
//@ assert *a == \at(*a,Pre);
  *a = *a +1;
//@ assert *a == \at(*a,Pre)+1;
  *a = *a +1;
//@ assert *a == \at(*a,Pre)+2;
  *b = *b +*a;
//@ assert *a == \at(*a,Pre)+2 && *b == \at(*b,Pre)+\at(*a,Pre)+2;
  return 0;
}
```

–

Example for $\backslash at(e, id)$

(autre label)

Listing 25 – project-divers/at2.c

```
void f (int n) {
for (int i = 0; i < n; i++) {
/*@ assert \at(i,LoopEntry) == 0; */
int j=0;
while (j++ < i) {
/*@ assert \at(j,LoopEntry) == 0; */
/*@ assert \at(j,LoopCurrent) + 1 == j; */
} }
}
```

Example for $\backslash at(e, id)$

(otherlabel)

Listing 26 – project-divers/change1.c

```
/*@ requires \valid(a) && *a >= 0;
  @ assigns *a;
  @ ensures   *a == \old(*a)+2 && \result == 0;
*/
int  change1(int *a)
{  int x = *a;
   x = x + 2;
   *a = x;
   return 0;
}
```

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Ghost variable

- A variable called *ghost* allows to model a computed value useful for stating a model property : the ghost variable is hidden for the computer but not for the model.
- It should not change the semantics of others variables and should not change the effective variables.

Wrong use of ghost variable

(Bug)

Listing 27 – project-divers/ghost2.c

```
int f (int x, int y) {
  //@ghost int z=x+y;
switch (x) {
case 0: return y;
//@ ghost case 1: z=y;
// above statement is correct.
//@ ghost case 2: { z++; break; }
// invalid, would bypass the non-ghost default
default: y++; }
return y; }

int g(int x) { //@ ghost int z=x;
if (x>0){return x;}
//@ ghost else { z++; return x; }
// invalid, would bypass the non-ghost return
return x+1; }
```

(Ghost variable)

Listing 28 – project-divers/ghost1.c

```
/*@ requires a >= 0 && b >= 0;
  ensures 0 <= \result;
  ensures \result < b;
  ensures \exists integer k; a == k * b + \result; */
int rem(int a, int b) {
  int r = a;
 /*@ ghost   int q=0;    */
  /*@
    loop invariant
    a == q * b + r &&
    r >= 0 &&  r <= a;
    loop assigns r;
    loop assigns q;
// loop variant r;
*/
  while (r >= b) {
    r = r - b;
/*@ ghost     q = q+1;    */
  };
  return r;
}
```

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Labels in ACSL

In **Frama-C** and the **ACSL** language, the expression :

```
\at(e, L)
```

evaluates an expression `e` in the memory state corresponding to a label `L`.

Main labels

- `Pre` : state at the beginning of the function
- `Post` : state at the end of the function
- `Here` : current state
- `Old` : initial value (often in `ensures`)
- `LoopEntry` : state before the loop
- `LoopCurrent` : start of the current iteration
- `Init` : initial state of the programme

Example with Old

```
void swap(int *a, int *b){

  int tmp;

  /*@
    requires a != \null && b != \null;
    assigns *a, *b;
    ensures *a == \at(*b,Old);
    ensures *b == \at(*a,Old);
  */

  tmp = *a;
  *a = *b;
  *b = tmp;
}
```

Why use \at

In proofs with the **WP** plugin, we must compare :

- the state **before** execution
- the state **after** execution

Example :

$$\cdot a_{final} = b_{initial}$$

We therefore write :

```
*a == \at(*b,Pre)
```

This allows the prover to reason about the **initial values of the programme**.

Example

Listing 29 – contrat

```
/*@ requires x >= 0;
@ ensures \result == \old(x) ;
@*/
int f(int x)
{
  int y;
  y=x;
  x=x+1;
  return(y);
 }
```

Example

Listing 30 – contrat

```
/*@ requires x >= 0;
@ ensures \result == \old(x) && \old(x) == x;
@*/
int f(int x)
{int y;
   y=x;
   x=x+1;
   /*@ assert x == \at(x,Pre)+1; */
   return(y);
}
```

Example

Listing 31 – contrat

```
/*@ requires x >= 0;
@ assigns \nothing;
@ ensures \result == \old(x);
@*/
int f(int x)
{int y;
  y=x;
  x=x+1;
  return(y);
}
```

Current Summary

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   - Second annotation
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Verification Condition for Assignment

Listing 32 – an1.c

```
//@ assert l1: P(x);
  x = e(x);
//@ assert l2: Q(x);
```

Verification Condition for Assignment

Listing 33 – an1.c

```
//@ assert l1: P(x);
  x = e(x);
//@ assert l2: Q(x);
```

- $P(x) \Rightarrow WP(x := e(x))(Q(x))$

Verification Condition for Assignment

Listing 34 – an1.c

```
//@ assert l1: P(x);
  x = e(x);
//@ assert l2: Q(x);
```

- $P(x) \Rightarrow WP(x := e(x))(Q(x))$
- $P(x) \Rightarrow Q[x \mapsto e(x))]$

Verification Condition for Assignment

Listing 35 – an1.c

```
//@ assert l1: P(x);
  x = e(x);
//@ assert l2: Q(x);
```

- $P(x) \Rightarrow WP(x := e(x))(Q(x))$
- $P(x) \Rightarrow Q[x \mapsto e(x))]$
- $P(x1) \Rightarrow Q[x \mapsto e(x1))]$ (renaming of free occurences of x by x1)

Verification Condition for Assignment

Listing 36 – an1.c

```
//@ assert l1: P(x);
  x = e(x);
//@ assert l2: Q(x);
```

- $P(x) \Rightarrow WP(x := e(x))(Q(x))$
- $P(x) \Rightarrow Q[x \mapsto e(x))]$
- $P(x1) \Rightarrow Q[x \mapsto e(x1))]$ (renaming of free occurences of x by x1)
- $P(x1) \wedge x = e(x1) \Rightarrow Q(x)$

Verification Condition for Assignment

Listing 37 – an1.c

```
//@ assert l1: P(x);
  x = e(x);
//@ assert l2: Q(x);
```

- $P(x) \Rightarrow WP(x := e(x))(Q(x))$
- $P(x) \Rightarrow Q[x \mapsto e(x))]$
- $P(x1) \Rightarrow Q[x \mapsto e(x1))]$ (renaming of free occurences of x by x1)
- $P(x1) \wedge x = e(x1) \Rightarrow Q(x)$
- $P(x1) \wedge x = e(x1) \Rightarrow Q(x)$

Verification Condition for Assignment

Listing 38 – an1.c

```
//@ assert l1: P(x);
  x = e(x);
//@ assert l2: Q(x);
```

- $P(x) \Rightarrow WP(x := e(x))(Q(x))$
- $P(x) \Rightarrow Q[x \mapsto e(x))]$
- $P(x1) \Rightarrow Q[x \mapsto e(x1))]$ (renaming of free occurences of x by x1)
- $P(x1) \wedge x = e(x1) \Rightarrow Q(x)$
- $P(x1) \wedge x = e(x1) \Rightarrow Q(x)$
- $\vdash P(x1) \wedge x = e(x1) \Rightarrow Q(x)$

Verification Condition for Assignment

Listing 39 – an1.c

```
//@ assert l1: P(x);
  x = e(x);
//@ assert l2: Q(x);
```

- $P(x) \Rightarrow WP(x := e(x))(Q(x))$
- $P(x) \Rightarrow Q[x \mapsto e(x))]$
- $P(x1) \Rightarrow Q[x \mapsto e(x1))]$ (renaming of free occurences of x by x1)
- $P(x1) \wedge x = e(x1) \Rightarrow Q(x)$
- $P(x1) \wedge x = e(x1) \Rightarrow Q(x)$
- $\vdash P(x1) \wedge x = e(x1) \Rightarrow Q(x)$
- $P(x1) \wedge x = e(x1) \vdash Q(x)$

Verification Condition for Assignment

Listing 40 – an1.c

```
//@ assert l1: P(x);
  x = e(x);
//@ assert l2: Q(x);
```

- $P(x) \Rightarrow WP(x := e(x))(Q(x))$
- $P(x) \Rightarrow Q[x \mapsto e(x))]$
- $P(x1) \Rightarrow Q[x \mapsto e(x1))]$ (renaming of free occurences of x by x1)
- $P(x1) \wedge x = e(x1) \Rightarrow Q(x)$
- $P(x1) \wedge x = e(x1) \Rightarrow Q(x)$
- $\vdash P(x1) \wedge x = e(x1) \Rightarrow Q(x)$
- $P(x1) \wedge x = e(x1) \vdash Q(x)$

  Assume $\{$
  $P(x1)$
- $x = e(x1)$
  $\}$
  Prove:$Q(x)$
-

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Annotation simple (I)

Listing 41 – an1.c

```
int main(void){
  signed long int x,y,z; //     int x,y,z;
  x = 1;
  /*@  assert x == 1; */
  y = 2;
  /*@ assert x == 1 && y == 2; */
  z = x *y;
  /*@  assert x == 1 && y == 1 && z == 2; */
  return 0;
  }
```

Annotation simple(I)

```
[kernel] Parsing an1.c (with preprocessing)
[wp] Running WP plugin...
[wp] Warning: Missing RTE guards
[wp] 2 goals scheduled
[wp] Proved goals:    4 / 4
  Terminating:     1
  Unreachable:     1
  Qed:             2
```

Annotation simple (I)

```
Goal Assertion 'l1' (file an1.c, line 3):
Assume {
  Type: is_sint32(x) /\ is_sint32(y).
  (* Initializer *)
  Init: x = 12.
  (* Initializer *)
  Init: y = 24.
}
Prove: (2 * x) = y.
Prover Qed returns Valid
```

Annotation simple (I)

```
------------------------------------------------------------

Goal Assertion 'l2' (file an1.c, line 5):
Assume {
  Type: is_sint32(x) /\ is_sint32(x_1) /\ is_sint32(y).
  (* Initializer *)
  Init: x_1 = 12.
  (* Initializer *)
  Init: y = 24.
  (* Assertion 'l1' *)
  Have: (2 * x_1) = y.
  Have: (1 + x_1) = x.
}
Prove: (2 + y) = (2 * x).
Prover Qed returns Valid
```

Annotation simple(I)

```
[kernel] Parsing an1.c (with preprocessing)
[wp] Running WP plugin...
[wp] Warning: Missing RTE guards
[wp] 2 goals scheduled
[wp] Proved goals:    4 / 4
  Terminating:     1
  Unreachable:     1
  Qed:             2
```

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Annotation simple (II)

Listing 42 – an2.c

```
void ex(void) {
  int x=12,y=24;
  //@ assert l1: 2*x == y;
  x = x+1;
  //@ assert l2:  y == 2*(x-1);
  x = x+2;
  //@ assert l3:  y+6 == 2*x;
}
```

Annotation simple (II)

```
[kernel] Parsing an2.c (with preprocessing)
[wp] Running WP plugin...
[wp] Warning: Missing RTE guards
[wp] 3 goals scheduled
[wp] Proved goals:    5 / 5
  Terminating:     1
  Unreachable:     1
  Qed:             3
```

Annotation simple (ii)

```
Goal Assertion 'l1' (file an2.c, line 3):
Assume {
  Type: is_sint32(x) /\ is_sint32(y).
  (* Initializer *)
  Init: x = 12.
  (* Initializer *)
  Init: y = 24.
}
Prove: (2 * x) = y.
Prover Qed returns Valid
```

Annotation simple (ii)

```
Goal Assertion 'l2' (file an2.c, line 5):
Assume {
  Type: is_sint32(x) /\ is_sint32(x_1) /\ is_sint32(y).
  (* Initializer *)
  Init: x_1 = 12.
  (* Initializer *)
  Init: y = 24.
  (* Assertion 'l1' *)
  Have: (2 * x_1) = y.
  Have: (1 + x_1) = x.
}
Prove: (2 + y) = (2 * x).
Prover Qed returns Valid
```

Annotation simple (ii)

```
Goal Assertion 'l3' (file an2.c, line 7):
Assume {
  Type: is_sint32(x) /\ is_sint32(x_1) /\ is_sint32(x_2) /\ is_s
  (* Initializer *)
  Init: x_2 = 12.
  (* Initializer *)
  Init: y = 24.
  (* Assertion 'l1' *)
  Have: (2 * x_2) = y.
  Have: (1 + x_2) = x_1.
  (* Assertion 'l2' *)
  Have: (2 + y) = (2 * x_1).
  Have: (2 + x_1) = x.
}
Prove: (6 + y) = (2 * x).
Prover Qed returns Valid
```

Listing 43 – an2bis.c

```
void ex(void) {
  int x=12,y=24;
  //@ assert l1: 2*x == y;
  x = x+1;
  //@ assert l2:  y == 2*(x-1);
   x = x+2;
  //@ assert l3:  y+6 == 2*x;
}
```

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Two Memory Models

- Hoare Model : `-wp hoare` is the option of frama-c
- Typed Model : default model is typed model

Hoare Model

- It simply maps each C variable to one pure logical variable.
- Heap cannot be represented in this model, and expressions such as *p cannot be translated at all.
- You can still represent pointer values, but you cannot read or write the heap through pointers.

Typed Model

- The default model for WP plug-in.
- Heap values are stored in several separated global arrays, one for each atomic type (integers, floats, pointers) and an additional one for memory allocation.
- Pointer values are translated into an index into these arrays.
- all C integer types are represented by mathematical integers and each pointer type to a given type is represented by a specific logical abstract datatype.

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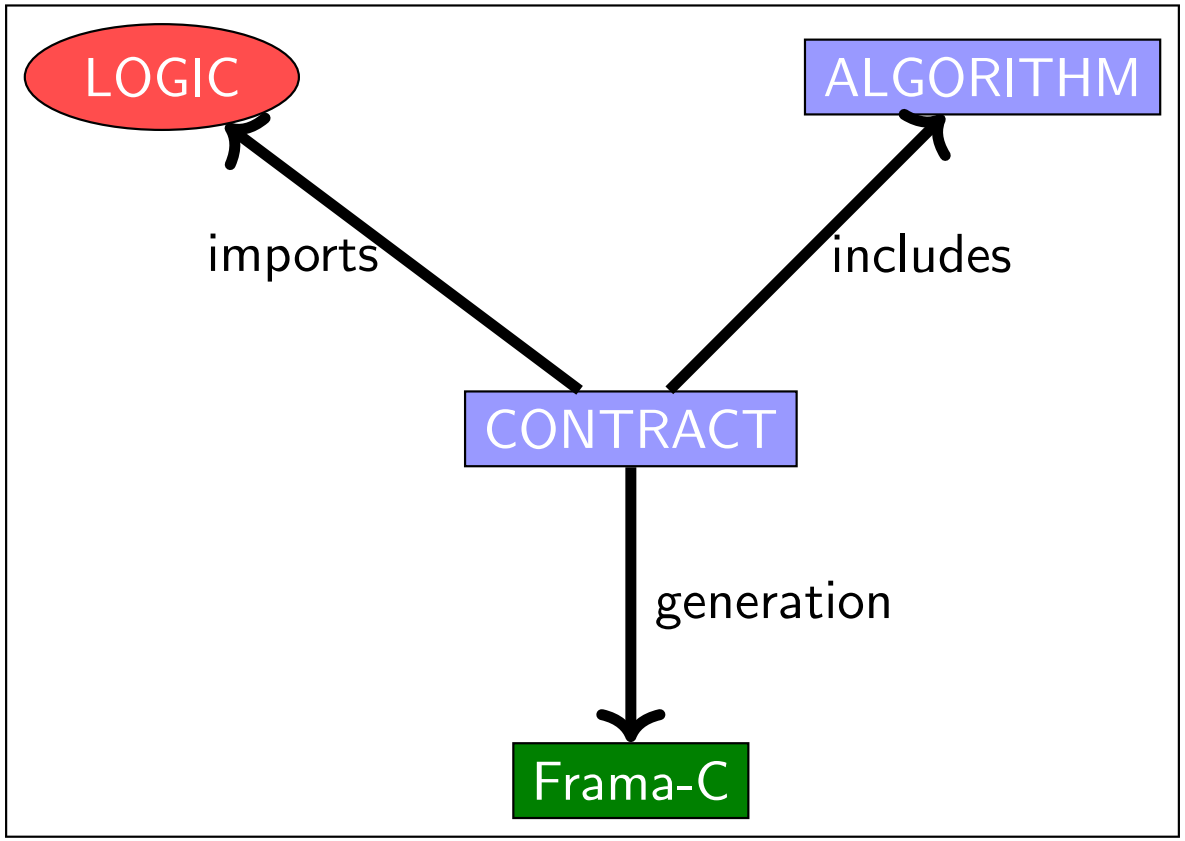
Tryptic Pattern



Defining domain properties in logical theory

- predicate

Predicates - Logic - Lemma (1)

(Predicate)

Listing 44 – project-divers/predicate1.c

```
/*@ predicate is_positive(integer x) = x > 0; */

/*@ logic integer get_sign(real x) =   @ x>0.0?1:(x<0.0?-1:0);
*/
/*@ logic integer max(int x,int y) =    x>=y?x:y;
*/
```

(Lemma)

Listing 45 – project-divers/lemma1.c

```
/*@ lemma div_mul_identity:
@ \forall real x, real y; y != 0.0 ==> y*(x/y) == x; @*/

/*@  lemma div_qr:
@ \forall int a, int b; a >= 0 && b >0 ==>
\exists  int q, int r; a == b*q +r && 0<=r && r <b; @*/
```

Predicates - Logic - Lemma (2)

(Definition of fibonacci function)

Listing 46 – project-divers/predicate2.c

```
/*@ axiomatic mathfibonacci{
  @ logic integer mathfib(integer n);
  @ axiom mathfib0: mathfib(0) == 1;
  @ axiom mathfib1: mathfib(1) == 1;
  @ axiom mathfibrec: \forall integer n; n > 1
  ==> mathfib(n) ==  mathfib(n-1)+mathfib(n-2);
  @ } */
```

(Definition of gcd)

Listing 47 – project-divers/predicate3.c

```
/*@ inductive is_gcd(integer a, integer b, integer d) {
@ case gcd_zero:
@ \forall integer n; is_gcd(n,0,n);
@ case gcd_succ:
@ \forall integer a,b,d; is_gcd(b, a % b, d) ==> is_gcd(a,b,d); @}
@*/
```

Predicates - Logic - Lemma (3)

(Definition of function odd/even)

Listing 48 – project-divers/predicate4.c

```
//@ predicate pair(integer x) = (x/2)*2==x;
//@ predicate impair(integer x) = (x/2)*2!=x;
//@ lemma ex: \forall integer a,b; a < b ==> 2*a < 2*b;

/*@ inductive is_gcd(integer a,integer b , integer c) {
  case zero: \forall integer n; is_gcd(n,0,n);
   case un: \forall integer u,v,w; u >= v ==> is_gcd(u-v,v,w);
   case deux: \forall integer u,v,w; u < v ==> is_gcd(u,v-u,w);
   }
*/
```

wp,wlp,loop ...

STATES l'ensemble des états sur l'ensemble X des variables, S une instruction de programme sur X, Q une propriété d'éttats, $s \in STATES$.

$$wp(S)(Q)(s) \stackrel{def}{=} \left(\begin{array}{l} \forall t \in STATES : \mathcal{D}(S)(s) = t \Rightarrow tQ(t) \\ \wedge \\ \exists t \in STATES : \mathcal{D}(S)(s) = t \end{array} \right) \ldots$$

wp,wlp,loop . . .

STATES l'ensemble des états sur l'ensemble X des variables, S une instruction de programme sur X, Q une propriété d'éttats, $s \in STATES$.

$$wp(S)(Q)(s) \stackrel{def}{=} \left(\begin{array}{l} \forall t \in STATES : \mathcal{D}(S)(s) = t \Rightarrow tQ(t) \\ \wedge \\ \exists t \in STATES : \mathcal{D}(S)(s) = t \end{array} \right) \quad \ldots \quad \ldots$$

$$wlp(S)(Q)(s) \stackrel{def}{=} (\forall t \in STATES : \mathcal{D}(S)(s) = t \Rightarrow Q(t))$$

$$\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots$$

$$wlp(S)(Q)(s) \equiv \overline{(\exists t \in STATES : \mathcal{D}(S)(s) = t \wedge \overline{Q}(t))} \quad \ldots\ldots\ldots$$

wp,wlp,loop ...

STATES l'ensemble des états sur l'ensemble X des variables, S une instruction de programme sur X, Q une propriété d'éttats, $s \in STATES$.

$$wp(S)(Q)(s) \stackrel{def}{=} \left(\begin{array}{l} \forall t \in STATES : \mathcal{D}(S)(s) = t \Rightarrow tQ(t) \\ \wedge \\ \exists t \in STATES : \mathcal{D}(S)(s) = t \end{array} \right) \quad \ldots \quad \ldots$$

$$wlp(S)(Q)(s) \stackrel{def}{=} (\forall t \in STATES : \mathcal{D}(S)(s) = t \Rightarrow Q(t))$$

$$wlp(S)(Q)(s) \equiv \overline{(\exists t \in STATES : \mathcal{D}(S)(s) = t \wedge \overline{Q}(t))} \quad \ldots\ldots \quad \ldots\ldots$$

- $loop(S) \equiv \overline{(\exists t \in STATES : \mathcal{D}(S)(s) = t}$ (ensemble des états qui ne permettent pas à S de terminer)
- $wp(S)(Q)(s) \equiv wlp(S)(Q)(s) \wedge \overline{loop(S)}(s)$

Weakest Liberal Precondition of S for P

$$\{S\}P(x) \stackrel{def}{=} \forall x_f . x \xrightarrow{\mathsf{S}} x_f \Rightarrow \mathsf{P}(x_f)$$

Logique de Hoare Correction Totale

⊠ Definitiontriplets de Hoare Correction Totale

$[P]\mathbf{S}[Q] \stackrel{def}{=} P \Rightarrow wp(S)(Q)$

Logique de Hoare Correction Totale

⊠ Definitiontriplets de Hoare Correction Totale

$[P]\mathbf{S}[Q] \stackrel{def}{=} P \Rightarrow wp(S)(Q)$

⊠ Definition(Axiomes et règles d'inférence)

- Axiome d'affectation : $[P(e/x)]$**X :=E(X)**$[P]$.
- Axiome du saut : $[P]$**skip**$[P]$.
- Règle de composition : Si $[P]\mathbf{S}_1[R]$ et $[R]\mathbf{S}_2[Q]$, alors $[P]\mathbf{S}_1$ **;**$\mathbf{S}_2[Q]$.
- Si $[P \wedge B]\mathbf{S}_1[Q]$ et $[P \wedge \neg B]\mathbf{S}_2[Q]$, alors $[P]$**if B then S**$_1$ **then S**$_2$ **fi**$[Q]$.
- Si $[P(n{+}1)]\mathbf{S}[P(n)]$, $P(n{+}1) \Rightarrow b$, $P(0) \Rightarrow \neg b$, alors $[\exists n \in \mathbb{N}.P(n)]$**while B do S od**$[P(0)]$.
- Règle de renforcement/affaiblissement : Si $P' \Rightarrow P$, $[P]\mathbf{S}[Q]$, $Q \Rightarrow Q'$, alors $[P']\mathbf{S}[Q']$.

Correction

:
Si $[P]$**S**$[Q]$ est dérivé selon les règles ci-dessus, alors $P \Rightarrow wp(S)Q)$.

- $[P(e/x)]$**X :=E(X)**$[P]$ est valide : $wp(X := E)(P)/x = P(e/x)$.
- $[\exists n \in \mathbb{N}.P(n)]$**while B do S od**$[P(0)]$: si s est un état de P(n) alors au bout de n boucles on atteint un état s_f tel que P(0) est vrai en s_f.

Complétude

:
Si $P \Rightarrow wp(S)(Q)$, alors il existe une preuve de $[P]\mathbf{S}[Q]$ construites avec les règles ci-dessus,

- $P \Rightarrow wp(X := E(X))(Q)$: $P \Rightarrow Q(e/x)$ et $[Q(e/x)]$**X :=E(X)**$[Q]$ constituent une preuve.
- $P \Rightarrow wp(while)(Q)$:
  - On construit la suite de $P(n)$ en définissant $P(n) = W_n$.
  - On vérifie que cela vérifie la règle du while.

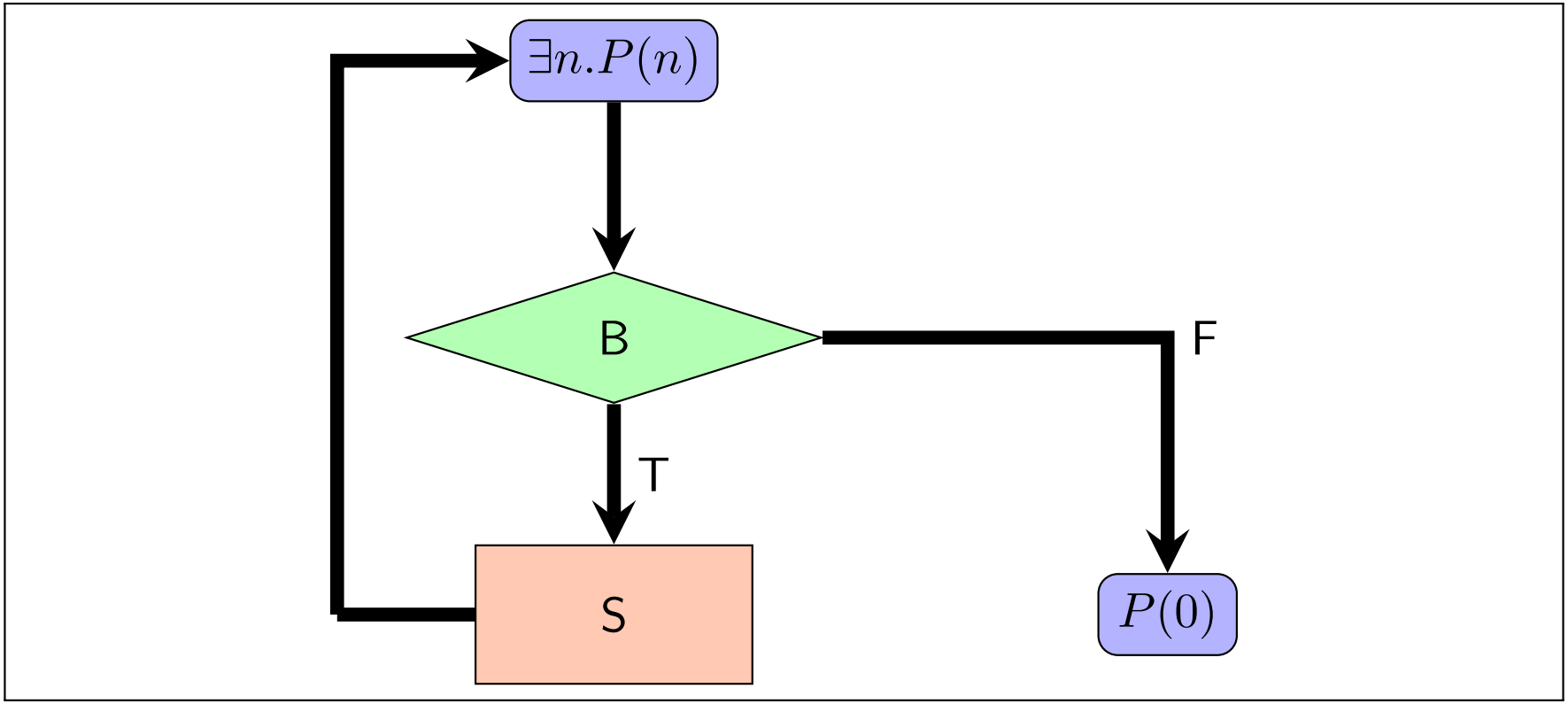
Introducing a variant

Introducing a variant

- assignment and skip are terminating
- If S1 and S2 are terminating, then S1 ;S2 is terminating and if(B,S1,S2) is terminating

Introducing a variant

- assignment and skip are terminating
- If S1 and S2 are terminating, then S1 ;S2 is terminating and if(B,S1,S2) is terminating



Explanation with the least fixed-point

- An assertion Q over $\mathbb{S}$ maps states of $\mathbb{S}$ to booleans of $\mathbb{B}$: $Q \in \mathbb{S} \longrightarrow \mathbb{B}$
- $\mathbb{A} \stackrel{def}{=} \mathbb{S} \longrightarrow \mathbb{B}$
- $\Rightarrow$ defines a partial ordering over $\mathbb{A}$
- $(\mathbb{A}, \Rightarrow)$ is a complete lattice with a smallest element $\perp$, a greatest element $\top$.
- For any Q in $\mathbb{A}$, $set(Q) = \{s \in \mathbb{S} | Q(s) = 1\}$ and $(\mathcal{P}(\mathbb{S}), \subseteq)$ is a complete lattice and $Q1 \Rightarrow Q2 \equiv set(Q1) \subseteq set(Q2)$
- $w(B)(S) \stackrel{def}{=}$ while B do S od
- $F_{B,S}(X) = B \wedge wp(S)(X) \vee \neg B \wedge Q$
- $F_{B,S}(\bigcup_{i \in I} X_i) = \bigcup_{i \in I} F_{B,S}(A_i)$ for any index sequence of assertions A_i $i \in I$.
- $F_{B,S}$ is a continuous map and has a least fixed-point by Kleene's theorem.
- $lfp(F_{B,S}) = \bigcup_{i \in \mathbb{N}} F^i_{B,S}$ with $F^0_{B,S} = \perp$ and $F^{i+1}_{B,S} = F_{B,S}(F^i_{B,S})$ for any $i \in \mathbb{N}$.

Defining approximations of while loop

- $F^0_{B,S} = \bot$: the first approximation means that the loop is diverging
- $F^{i+1}_{B,S} = B \wedge wp(S)(F^i_{B,S}) \vee \neg B \wedge Q$
- $P(0) \stackrel{def}{=} F^1_{B,S} \equiv \neg B \wedge Q$
- $\ldots$
- $P(i{+}1) \stackrel{def}{=} B \wedge wp(S)(P(i))$

The sequence of assertions $P(i)$ satisfies the following properties

- $\forall i \in \mathbb{N}, [P(i{+}1]S[P(i)]$
- $\forall i \in \mathbb{N}, [P(i{+}1] \rightarrow B$
- $P(0) \Rightarrow \neg B$

By the rule of Hoare Logic for Total Correctness, we get the following property $[\exists i \in \mathbb{N}, P(i]w(B)(S)[P(0)]$.

Loop termination

- The termination is proved by showing that each loop terminates.
- Any loop is characterized by an expression `expvariant(x)` called `variant` which should decrease each execution of the body :
  $\forall x_1, x_2. b(x_1) \wedge x_1 \xrightarrow{S} x_2 \Rightarrow \text{expvariant}(x_1) > \text{expvariant}(x_2)$

(Variant)

Listing 49 – project-divers/variant1.c

```
/*@ requires n > 0;
  terminates n > 0;

  ensures \result == 0;
*/
int code(int n) {
  int x = n;
  /*@ loop invariant x >= 0 &&  x <= n;
    loop assigns x;
    loop variant x;
   */
  while (x != 0) {
    x = x - 1;
  };
  return x;
}
```

Example of loop variant

(Variant)

Listing 50 – project-divers/variant3.c

```
int f() {
int x = 0;
int y = 10;
/*@
    loop invariant
    0 <= x < 11 && x+y == 10;
    loop variant y;
   */
while (y > 0) {
  x++;
  y--;
}
 return 0;
}
```

Recursive Call

(Variant)

Listing 51 – project-divers/variant4.c

```
g/*@ requires n <= 12;
  @ decreases n;
  @*/
int fact(int n){
  if (n <= 1) return 1;
  return n*fact(n-1);
}
```

Terminates

(Terminates)

Listing 52 – project-divers/terminates1.c

```
/*@
  requires P;
  terminates Q;
*/
void f(...);
```

- Frama-C génère la condition de vérification VC terminates
- $P \Rightarrow Q$

- – lemma : 1 VC
- – axiom : no VC (admitted with no proof)
- – ensures : 1 VC
- – exits : 1 VC
- – disjoint : 1 VC
- – complete : 1 VC
- – requires : 1 VC for each call
- – terminates : 1 VC for each call, 1 VC for each loop without "loop variant"
- – decreases : 1 VC for each recursive call
- – assigns : 1 VC for each assigned lvalue
- – admit : no VC (admitted with no proof)
- – assert/check : 1 VC
- – loop invariant : 2 VCs (established, preserved)
- – loop variant (integer) : 2 VCs (positive, decreasing)
- – loop variant (general measure) : 1 VC (the measure is assumed to be well-founded) – loop assigns : 1 VC for each assigned lvalue within the loop

Current Summary

1. Contracts
   - Extending C programming language by contracts
   - Playing with variables
   - Ghost Variables
   - Labels
2. Generation of Verification Conditions
   - WP calculus in Frama-c
   - First annotation
   - Second annotation
3. Memory Models in Frama-c
4. Logic Specification
5. Organisation of the verification process
6. Conclusion

Main steps

- Defining th mathematical function to compute *mathf*
- Stating the postcondition using the mathematical function
- Evaluating the inductive sequence u_i computing the function *mathf*
- $\forall i \in \mathbb{N} : u_i = marthf(i)$
- Evaluating relationship among variables.

The power two function

(power2.h)

Listing 53 – project-powers/power21.h

```
#ifndef _A_H
#define _A_H
// Definition of the mathematical function mathpower2
/*@ axiomatic mathpower2 {
  @ logic integer mathpower2(integer n);
  @ axiom mathpower2_0: mathpower2(0) == 0;
  @ axiom mathpower2_rec: \forall integer n; n > 0
  ==> mathpower2(n) == mathpower2(n-1) + n+n+1;
  @ } */
/*@ axiomatic matheven {
  @ logic integer matheven(integer n);
  @ axiom matheven_0: matheven(0) == 0;
  @ axiom mathpeven_rec: \forall integer n; n > 0
  ==> matheven(n) == matheven(n-1) + 2;
  @ } */
// We define v and w in a one shot axiomatic definition
/*@ axiomatic vw {
  @ logic integer v(integer n);
  @ logic integer w(integer n);
  @ axiom v_0: v(0) == 0;
  @ axiom w_0: w(0) == 0;
  @ axiom v_rec: \forall integer n; n > 0
  ==> v(n) == v(n-1) +n+n+1 && w(n) == w(n-1) + 2;
@ } */
```

The power two function

(power2.h)

Listing 54 – project-powers/power22.h

```
/*@ lemma propw:
@ \forall int n; n >= 0 ==> w(n) == n+n; @*/
/*@ lemma propv:
@ \forall int n; n >= 0 ==> v(n) == n*n; @*/
/*@ lemma prop1:
@ \forall int n; n >= 0 ==> matheven(n) == n+n; @*/
/*@ lemma prop2:
@ \forall int n; n >= 0 ==> mathpower2(n) == n*n; @*/
/*@ axiomatic auxmath {
  @ lemma rule1: \forall int n; n >0 ==> n*n == (n-1)*(n-1)+2*(n-1)+1;
  @ } */
/*@  requires  0 <= x;
     assigns \nothing;
     ensures \result == x*x;
*/
int power2(int x);
#endif
```

Factorial function

(power2.h)

Listing 55 – project-powers/power2.c

```
#include <limits.h>
#include "power2.h"

int power2(int x)
{int r,k,cv,cw,or,ok,ocv,ocw;
  r=0;k=0;cv=0;cw=0;or=0;ok=k;ocv=cv;ocw=cw;
      /*@ loop invariant 0 <= cv && 0 <= cw && 0 <= k;
        @ loop invariant cv == k*k;
         @ loop invariant k <= x;
         @ loop invariant cw == 2*k;
         @ loop invariant 4*cv == cw*cw;
         @ loop assigns k,cv,cw,or,ok,ocv,ocw;
        @ loop variant x-k;
*/
  while (k<x)
    {
    ok=k;ocv=cv;ocw=cw;
      k=ok+1;
      cv=ocv+ocw+1;
      cw=ocw+2;

    }
  r=cv;
  return(r);
}
```

Current Summary

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Principles for RTE

- Plugin -rte adds specifi assertions for each variable
- Systematic checking of RTE.

- Defining domain properties (axioms, lemmas, proofs)
- Defining loop invariants (typing, equation, ...)
- Analyzing inductive properties
- Identifying inputs (*requires*) and outputs (*ensures*)